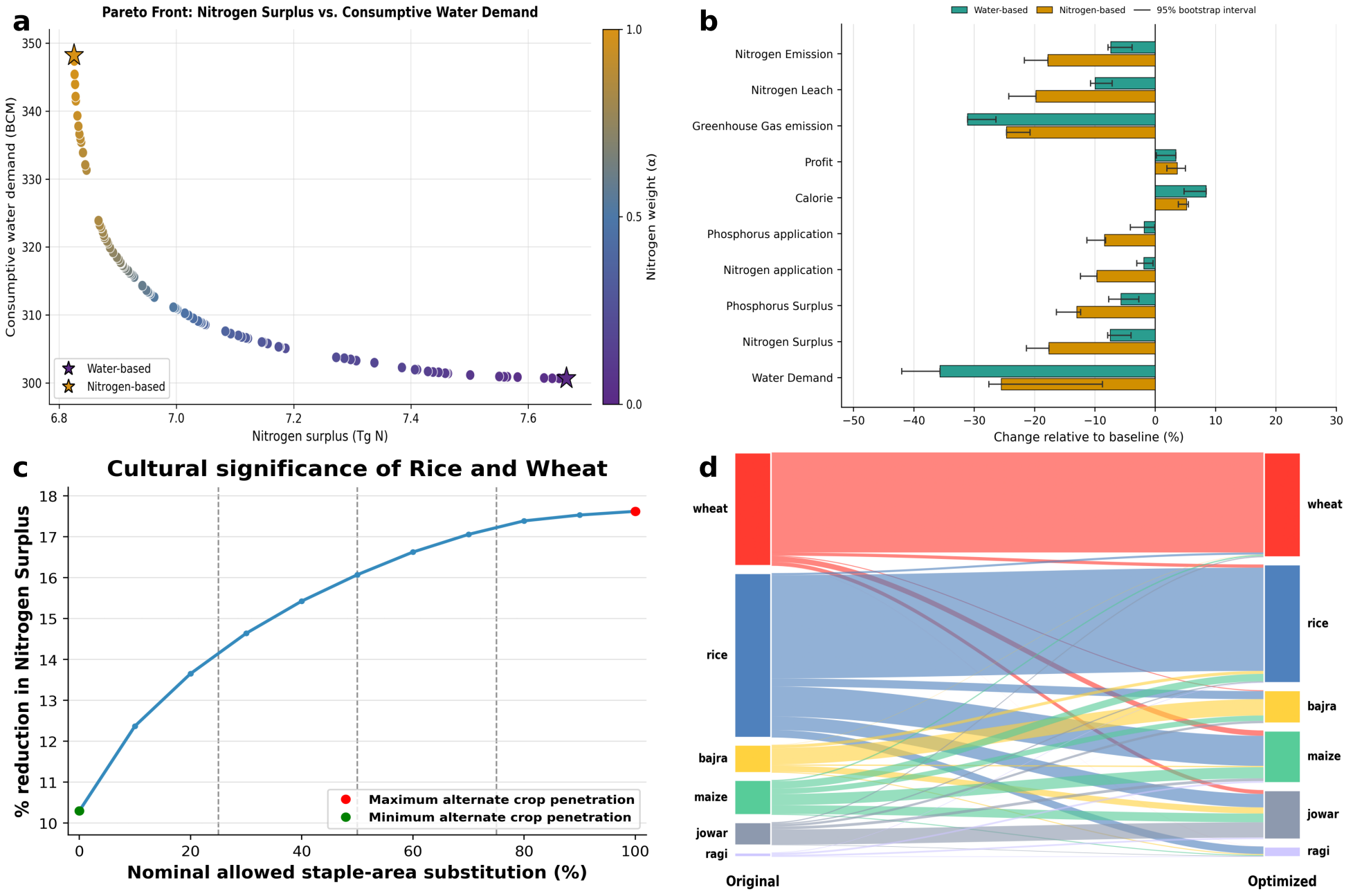
a
Pareto Front: Nitrogen Surplus vs. Consumptive Water Demand
Consumptive water demand (BCM)
Nitrogen surplus (Tg N)
Nitrogen weight (α)
Water-based
Nitrogen-based
b
Water-based
Nitrogen-based
95% bootstrap interval
Nitrogen Emission
Nitrogen Leach
Greenhouse Gas emission
Profit
Calorie
Phosphorus application
Nitrogen application
Phosphorus Surplus
Nitrogen Surplus
Water Demand
Change relative to baseline (%)
c
Cultural significance of Rice and Wheat
% reduction in Nitrogen Surplus
Nominal allowed staple-area substitution (%)
Maximum alternate crop penetration
Minimum alternate crop penetration
d
wheat
rice
bajra
maize
jowar
ragi
Original
Optimized